Digital Harmonium Workstation

Phase 3.1 — System Design

Meeting 3 — User Interface, Navigation, and Human Interaction

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Meeting Objective

The purpose of this meeting was to completely define how a musician interacts with the Digital Harmonium Workstation.

Unlike previous meetings that focused on project identity and hardware features, this meeting focused entirely on the human experience.

The objective was to answer one fundamental question:

> **When a musician sits in front of the finished instrument, what exactly happens from the moment power is applied until music is being performed?**

Every screen, button, navigation path, interaction philosophy, and display decision discussed in this meeting becomes part of the permanent system architecture.

Core Design Philosophy

The user interface shall behave like a musical instrument rather than a computer application.

The musician should never feel like they are navigating software.

Instead, interaction should feel immediate, simple, and predictable.

Menus should remain shallow.

Performance-related controls should always require the smallest possible number of button presses.

The instrument should disappear into the musical experience.

UI Design Principles

The following principles were agreed upon.

Principle 1

The instrument is designed for musicians, not programmers.

The interface must therefore prioritize speed, clarity, and simplicity.

Principle 2

The LCD should never become cluttered.

Only information relevant to the current task should be displayed.

Principle 3

Menus should remain extremely shallow.

Deep hierarchical navigation is intentionally avoided.

Principle 4

Frequently used musical functions should always remain directly accessible.

A musician should never have to navigate through multiple menus during performance.

Principle 5

The interface must remain usable without reading a manual.

A first-time user should quickly understand how to operate the instrument.

Boot Sequence

A simple splash screen alone was considered insufficient.

Instead, the boot process itself should perform useful initialization and hardware verification.

The agreed boot sequence is:

1. Power applied.
2. ESP32 boots.
3. LCD initializes.
4. Hardware self-test begins.
5. LEDs perform a brief startup sweep.
6. Audio subsystem produces a short startup confirmation tone.
7. Keyboard matrix initializes.
8. SD card initializes.
9. Internal memory and flash initialize.
10. Default instrument is loaded.
11. Home screen appears.

Boot Display

Possible display sequence:

```
Digital Harmonium
Booting...
```

↓

```
Initializing...
```

↓

```
Loading Instrument...
```

↓

Home Screen.

The total startup duration should remain approximately three to five seconds.

The boot sequence is intended both for user feedback and as a diagnostic mechanism.

Display Constraints

An important design correction was made during this meeting.

Originally, a richer display layout had been considered.

However, the final hardware currently uses a **16×2 LCD**.

This fundamentally changes the interface philosophy.

The display cannot show large amounts of simultaneous information.

Instead,

every screen should only display information relevant to its own context.

This became one of the most important design decisions of the meeting.

Home Screen Philosophy

The Home Screen should remain intentionally minimal.

Its purpose is only to indicate the currently selected instrument and the general operating state.

It is not intended to become an information dashboard.

The Home Screen should immediately allow the musician to begin playing.

Context-Based Screens

Instead of placing every parameter on one display, each major subsystem receives its own dedicated screen.

Examples include:

- Voice
- Tabla
- Tanpura
- Metronome
- Recording
- Playback

- Settings

Each screen owns only its own parameters.

This keeps every display clean and understandable.

Voice Screen

The Voice screen allows cycling through available instruments.

Examples include:

- Harmonium
- Organ
- Flute (future)

The currently selected instrument is always displayed.

Selecting a voice immediately activates it.

No confirmation dialog is required.

Voice Information

Additional information may be displayed when useful, such as:

- Root sample key
- Active effects
- Future instrument metadata

Only information directly related to the selected voice should appear.

Navigation Model

A major improvement introduced during this meeting is the addition of a dedicated **Select** button.

Navigation now consists of two stages:

Stage 1

Cycle through available pages using dedicated buttons.

For example:

Voice button

↓

Voice page

Tabla button

↓

Tabla page

Tanpura button

↓

Tanpura page

Stage 2

Press the **Select** button to enter the detailed settings of the currently selected page.

This avoids unnecessary nested menus while still allowing access to advanced settings.

Tabla Screen

The Tabla page displays the currently selected taal.

Examples:

- Tritaal
- Ektaal
- Jhaptaal
- Rupak
- Dadra
- Chowtal

Pressing the Tabla button cycles through taals.

Pressing the Select button opens the detailed Tabla settings.

Possible settings include:

- Tempo
- Transpose
- Future rhythm options

Tanpura Screen

The Tanpura page displays:

- ON/OFF state

Selecting the page opens:

- Tempo
- Transpose
- Scale

Each parameter is edited separately using the same interface philosophy.

Tempo Philosophy

A significant architectural decision was made.

Tempo buttons should **not** always control the same subsystem.

Instead,

Tempo buttons operate on the currently selected context.

Examples:

If currently editing Tabla,

Tempo buttons modify Tabla tempo.

If editing Metronome,

Tempo buttons modify Metronome tempo.

If editing Tanpura,

Tempo buttons modify Tanpura timing.

This context-sensitive behavior significantly reduces interface complexity.

Transpose Philosophy

The same context-sensitive philosophy applies to transpose.

When editing an instrument,

Transpose modifies the currently selected voice.

When editing Tabla,

Transpose modifies Tabla pitch.

When editing Tanpura,

Transpose modifies Tanpura scale.

Recording Screen

Recording intentionally remains extremely simple.

The display should primarily show elapsed recording time.

Example:

```
Recording

00:15
```

No unnecessary information should appear during recording.

The recording subsystem stores musical events rather than audio.

Playback Screen

Playback follows the same minimalist philosophy.

The user first selects a recording.

During playback the display shows only playback status and elapsed time.

Example:

```
Playback

00:08
```

Educational LED Playback

One of the unique features finalized during this meeting is educational playback.

Instead of merely replaying recorded notes,

the instrument should illuminate the corresponding keyboard LEDs during playback.

This allows a student to visually follow exactly which key was played.

This transforms playback into a learning tool.

The LEDs effectively become a digital teacher.

This feature is considered one of the signature educational capabilities of the project.

Sliders

The original Casio keyboard already contains analog sliders.

Rather than adding additional hardware,

these sliders should be reused.

Current agreement:

Slider 1

Dedicated Master Volume.

Always available.

Slider 2

Reserved for future configurable functions.

Possible assignments include:

- Reverb amount
- Chorus amount
- Display contrast
- Effect depth
- Other future parameters

The exact function remains open.

Factory Reset

Factory Reset should reset only:

- Settings
- Configuration

It must **never** erase:

- SD card
- Recorded performances
- Instrument files

This distinction was explicitly agreed.

Displaying Musical Notes

Displaying the currently played note was discussed.

The conclusion was that this information provides little practical value during performance.

Constantly updating note names would create unnecessary distraction.

Therefore,

live note display will **not** be included in the normal interface.

It may exist only in future diagnostic or debugging modes.

Root Sample Information

Displaying the root key of the currently selected instrument was considered useful.

Possible example:

```
Harmonium

Root C4
```

This information is useful for developers and advanced users while remaining unobtrusive.

Status Indicators

Near the end of the meeting another improvement was proposed.

The Home Screen should display small status indicators showing important background subsystems.

Possible indicators include:

- Tanpura ON/OFF
- Tabla ON/OFF
- Recording active
- Playback active

These indicators may appear as compact text on the LCD.

An alternative future implementation is to dedicate one or more keyboard LEDs as persistent status indicators.

The exact implementation remains open, but the concept has been accepted.

Final User Interface Philosophy

The finalized philosophy can be summarized as follows:

- Boot quickly while verifying hardware.

- Keep every screen context-specific.
- Never overload the 16×2 LCD.
- Provide direct access to musical functions.
- Reserve menus only for detailed settings.
- Use the Select button to open contextual options.
- Reuse existing hardware wherever possible.
- Use LEDs not only for visual effects but also for education.
- Keep the interface musician-centric rather than software-centric.

Decisions Finalized During Meeting 3

✓ Hardware self-test integrated into boot sequence.

✓ Interface redesigned specifically for a 16×2 LCD.

✓ Context-based screens replace a single overloaded Home Screen.

✓ Dedicated Select button adopted for entering page-specific settings.

✓ Context-sensitive Tempo and Transpose controls adopted.

✓ Separate screens established for Voice, Tabla, Tanpura, Recording, Playback, and Settings.

✓ Recording and Playback interfaces simplified.

✓ Educational LED playback approved as a core feature.

✓ Existing Casio sliders reused rather than adding unnecessary controls.

✓ Factory Reset limited to system settings only.

✓ Live musical note display removed from normal operation.

✓ Root sample information accepted as optional display content.

✓ Compact subsystem status indicators accepted for future implementation.

Meeting 3 Status: COMPLETE

The complete human interaction model of the Digital Harmonium Workstation has now been defined.

Future runtime architecture, state machines, and software implementation will directly follow the interaction model established during this meeting.